

Metric Category	Detail	Value/Notes
Total Runtime	Application runtime	~2,016 ms per rank (varies slightly)
MPI Init Calls	Number	1 call per rank
MPI Finalize Calls	Number	1 call per rank
MPI Sendrecv	Count	~1302 calls per rank
	Avg Message Size	~13,000 bytes
	Message size range	~8,000 to 39,000 bytes
	Neighbors communication counts	221 to 430 calls per neighbor
MPI Allreduce	Count	40 calls per rank
	Message size	4-8 bytes (1 double)
	Average latency	~2450 us
MPI Barrier	Count	15 calls per rank
	Average latency	~1390 us
MPI Collective Sync	Collective sync events	Present, minor overhead
Mesh Size	Dimensions	2x2x4 (16 ranks/nodes)
Neighbor Ranks	Example neighbors for rank 0	Ranks 1, 3, 4, 8
Message size stats	Per neighbor	Around ~20,000 bytes and ~9,000 bytes